College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 02

Focus: Scanner I/O, operators, methods, fields, getters/setters intro

Difficulty: Easy

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Console input with Scanner.

a. Write the import and the line that creates a `Scanner` reading from `System.in`. **(4 marks)**

b. Write code that asks for a student's name and age, reads both, and prints them. **(8 marks)**

c. Which method reads a whole line of text? Which is commonly used for an `int`? **(4 marks)**

d. What problem can occur if the user types letters when an integer is expected? (Name the issue in plain words.) **(4 marks)**

Question 2 [20 Marks]

Operators and expressions.

a. Evaluate: `10 + 3 * 2` and `(10 + 3) * 2`. Show working. **(4 marks)**

b. Write a boolean expression that is true when `age` is at least 18 and `hasId` is true. **(4 marks)**

c. Explain the difference between `=` and `==` with a tiny example. **(6 marks)**

d. What does the modulus operator `%` return? Give one example. **(6 marks)**

Question 3 [20 Marks]

```
public static int add(int a, int b) {
    return a + b;
}
```

Study the method.

a. What are the return type and the two parameters? **(4 marks)**

b. Write a complete `main` that calls `add(7, 5)` and prints the result. **(6 marks)**

c. Rewrite the method as `void` so it prints the sum instead of returning it. **(6 marks)**

d. Can a `void` method use `return 5;`? Explain briefly. **(4 marks)**

Question 4 [20 Marks]

Fields vs local variables (simple).

a. Where is a field declared? Where is a local variable declared? **(6 marks)**

b. Write a class `Lamp` with a field `boolean isOn` and methods `turnOn()` / `turnOff()` that change it. **(8 marks)**

c. Explain why a local variable declared inside `turnOn` cannot be used inside `turnOff`. **(6 marks)**

Question 5 [20 Marks]

Getters and setters - first practice.

a. What is a getter used for? What is a setter used for? **(4 marks)**

b. For a private field `String title`, write `getTitle` and `setTitle`. **(8 marks)**

c. Improve `setTitle` so it rejects an empty string (print a message and keep the old value). **(6 marks)**

d. State the usual JavaBeans name for a getter of field `author`. **(2 marks)**

Question 6 [20 Marks]

Procedural vs OOP (simple comparison).

a. Describe procedural programming in two sentences. **(4 marks)**

b. Describe object-oriented programming in two sentences. **(4 marks)**

c. Name the four pillars of OOP. **(4 marks)**

d. Give one advantage of modelling a library with `Book` objects instead of only separate variables in `main`. **(8 marks)**

Question 7 [20 Marks]

Simple class design.

a. Using the noun/verb idea, list two fields and two methods for a `Student` class. **(6 marks)**

b. Write the Java class with those private fields and public methods (stubs are enough for methods that only print). **(10 marks)**

c. In `main`, create two different `Student` objects to show that each has its own data. **(4 marks)**

******* END OF QUESTION PAPER *******